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Design criteria for the optimal sizing of integrated photovoltaic-storage systems

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A B S T R A C T

The paper aims at presenting a methodology to choose the proper sizing of an integrated Photovoltaic unit and a storage architecture starting from the knowledge of the load and solar irradiance time profiles. To do this, first a simple logic to manage the charging/discharging of the storage is proposed, then the function representing the overall cost during the whole life of the system is derived. The computation of the cost function for different photovoltaic and storage sizes allows finding the best sizing of the system. Finally, a thorough economic analysis is conducted to provide criteria to check whether the investment is economically advantageous. This is done considering the impact of the economic variables which are more influential on the investment key assessment parameters. The proposed methodology has been validated in different realistic test cases against the results provided by an energy management system, showing an excellent agreement between the approaches. Moreover, the economic analysis has been applied to four realistic situations in order to tailor the microgrid components ratings on the specific test case.

1. Introduction

In the last few years, Photovoltaic (PV) generating units have experienced an extraordinary growth all over the world. As reported in Ref. [1], 2016 was a record year for PV power with an overall increase of globally installed power of 76.6 GW over a total of 306.5 GW at the end of 2016 [1]. China is the country that scored the major contribution to the increasing of PV installed power plants of 2016 with 34.5 GW, roughly 50% of the total increase. This increasing trend is expected to continue in the future with a forecast production of 500 GW for 2020 [1].

This evolution of the electric production mix has changed the operating conditions of the power system, determining benefits, new problems and important challenges for Distribution System Operators (DSOs). The need to mitigate the impact of Distributed Energy Resources (DER) on distribution networks and the level of "market maturity" [2] reached in several countries by technologies such as PV generation have accelerated the trend to "grid parity" for renewable generation; i.e. lower feed-in tariffs [3] and higher difference between purchasing and selling prices [4]. This, on the other hand, has increased the number of PV installations for self-consumption purposes, e.g. PV system designed to satisfy the local load with or without out the connection to the distribution network. Due to the stochastic behavior of PV systems, this configuration is sustainable only if an integrated PV-Storage solution is considered, in order to distribute the daytime energy production over the whole day.

The development of effective integration strategies for solar generating units is a need shared by many organizations and institutions all over the world and this is witnessed by the relevant number of initiatives to support and finance research programmes with the aim of making this PV-Storage integration feasible at industrial level. In the US the D.o.E. (Department of Energy) is supporting the SunShot Initiative programme with the goal of making the PV source competitive with other sources of energy supply by the end of the decade, achieving the integration of 10 GW of installed PV plants in the American electricity network [5]. In Saudi Arabia the most ambitious programme for the support and

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development of the solar energy source (KA-CARE Program) is currently ongoing, carried on by the King Abdullah University of Science and Technology (KAUST) of Thuwal [6]. Also the European Union, inside the Horizon 2020 programme, is promoting and supporting research activities aimed at increasing the flexibility of the energy system by means of the improvement and integration of energy storage systems (Low Carbon Energy—WP10 Secure, clean and efficient Energy Programme, LCE 8, 9 e 10) [7].

The common aspect of all the above-mentioned initiatives is the necessity to go beyond the sole technical aspects related to the integration such as suitable energy management strategies, communications between devices and so on, and focus on the economic sustainability of the integrated solution providing criteria to quantify the sustainability of the investments and defining the actions and targets to get this configuration competitive with traditional power generation.

From the technical point of view integrated storage solutions are applied in various contexts with different aims. The first one is maximizing the PV integration into the power system and improving the correlation with forecasted power. In particular, some papers have proposed the application of storage systems to increase the network hosting capacity for renewables e.g. Ref. [8] where a methodology has been proposed to allow the planner to decide the best compromise between cost of network upgrading, cost of power losses, cost of not supplied energy and cost of energy required by the served customers. On the other hand, [9] proposes a strategy to enable the minimization of the power peaks together with an analytical design method to achieve an optimal peak shaving. Storage application has also been applied in fuelled power systems [10]. In Ref. [11] a probabilistic method has been developed to predict the ability of energy storage to increase the penetration of intermittent embedded renewable generation on weak electricity grids and to enhance the value of the electricity generated by time-shifting delivery to the network. Another subject extensively treated in literature concerns the use of optimization methods for the sizing of renewables. For example, in Ref. [12] sizing and control methodologies for a zinc-bromine flow battery-based energy storage system have been presented, while in Ref. [13] the authors calculate the sizing of a hybrid energy system using various energy management strategies. In Ref. [14] a size optimization method of the energy sources in a standalone PV system is proposed. In Ref. [15] the authors present an optimization model based on integer programming for the adoption of stand-alone PV systems in the residential sector, which determines the optimal number of PV modules and batteries but also assesses the economic feasibility of the system. Finally, in Ref. [16] a multi-agent system based distributed Energy Management System (EMS) is proposed to perform an optimal energy allocation and management for grids consisting of renewables, storage and distributed generation.

As a result, manufacturers are proposing a number of solutions like the Sirio-Power-Supply (SPS) [17] and the Home-Business-Seiter [18], targeted also on residential applications, typically composed by a power converter plus a storage system, to be used in conjunction with renewable generation and eventually gensets.

From an economic point of view, the attention of many researchers is now also focusing on establishing criteria to assess the economic viability of such plants. This is the reason why the literature has recently grown with many works in this direction. Many authors consider the main economic analysis parameters to evaluate the system's economic sustainability. In Ref. [19] the authors analyse different economic parameters in order to determine the number of PV plants necessary to achieve a pre-set production goal with this type of plant in Italy, while in Ref. [20] an economic analysis is conducted for an investment in a PV plant located in a semi-desert area of Colorado. In Refs. [21] and [22], an economic analysis of various PV plants has been performed. Other authors, instead, analyse particular economic aspects. For example, in Ref. [23] the non-sustainability of investments in PV plants with decreased incentive rates is pointed out, while in Ref. [24] the authors focus on identifying a number of variables that might explain the variations in a solar plant's investment cost and then define in which areas of Chile an installation would be most advantageous. In Ref. [25] a comparative economic analysis is performed about the key policies supporting the dissemination of energy production by means of PV plants in several European countries, while in Ref. [26], the importance of the government incentives or the key economic policies/strategies are considered.

However, to the best of the authors knowledge, a simple but effective procedure for the optimal sizing of an integrated PV-Storage solution based on an extensive economic analysis is still missing. Therefore, the aim of the present article is that of filling this gap presenting a procedure that, starting from the solar irradiance and the electric load estimation, provides the best rating of both components. In accordance with the aim of the various financial support initiatives the focus is the economic sustainability of the integrated solution. This is done analysing all the classical economic indexes which are most influential on the investment key assessment parameters. In particular, Net Present Value (*NPV*), Internal Rate of Return (*IRR*), Pay Back Period (*PBP*) and Discounted Profitability Ratio (*DPR*) are taken into account.

This procedure requires the necessity of considering variable costs and revenues that are strictly related to the specific management strategy used to define the storage production. From a mathematical point of view, these algorithms result in optimization problems that are often cumbersome from a computational point of view as they are designed for much more complicated structures, in which network constraints have to be taken into account (see Refs. [27] and [28]). A structure like this cannot be effectively included into an overall economic evaluation of the system. Due to the necessity of keeping the economic assessment as simple as possible, a simplified storage management logic is introduced, namely the Heuristic Rule System (HRS), to be included in the optimal design procedure. Before evaluating the performances of the proposed sizing procedure in a realistic test case, a validation of the proposed HRS for the storage management is reported against a more sophisticated structure of energy management.

The paper is organized as follows: Section 2 includes the formulation of the proposed sizing criterion including the description of the HRS storage management logic. Then, Section 3 is devoted to validate the HRS comparing it with a traditional EMS and to present some realistic case studies; finally, some conclusions are drawn in Section 4.

2. Problem formulation

In this section, the proposed methodology for the choice of the rating of a PV unit integrated with a storage device is presented together with a simple but effective storage management strategy.

2.1. Nomenclature

In this part, a list of all the variables and symbols involved in the problem formulation is presented:

- N_t are the time intervals in the considered horizon (t denotes the generic time instant and Δt is the length of each time interval);

- $P_{PV,max}$ is the PV system rated active power;
- $P_{PV,t}$ is the active power of the PV system at time t;
- $P_{EL,t}$ is the electric load active power request at time t;
- $P_{NET,t}$ is the active power injected by the external network at time t;
- $\eta_{ST,in}$ and $\eta_{ST,out}$ represent the storage charging and discharging efficiency;
- $P_{ST,disch}$ and $P_{ST,ch}$ are the storage maximum power for the discharging and charging phase;
- $P_{ST,t}$, is the storage injected active power at time t;
- $W_{ST,t}$ is the storage energy content or State of Charge (SoC) at time t;
- $W_{ST,max}$ is the storage rated energy;
- $W_{ST,min}$ is the minimum storage SoC;
- C_{ST} is the cost of the storage system per installed kWh;
- C_{PV} is the PV cost per installed peak kW;
- $C_{sell,t}$ and $C_{buy,t}$ are the costs of the sold and purchased energy to/from the distribution grid;

2.2. Electric system and components representation

The electric system under consideration accounts for a PV unit, a storage device, an electric load and the interconnection to the external distribution grid as depicted in Fig. 1. For the future derivation, the PV unit can be fully described by the active power $P_{PV,t}$ produced at time step t, while the electric load is characterized by the power absorption $P_{EL,t}$ at the same time instant. The external network is supposed to be able to compensate any mismatch between the overall power absorption and production; in other words, the distribution system active power $P_{NET,t}$ can assume any positive or negative value.

The storage system is represented by the injected active power $P_{ST,t}$, while the storage energy $W_{ST,t}$ can be calculated as [12]:

$$\begin{cases} W_{ST,t} = W_{ST,t-1} - \eta_{ST,in} P_{ST,t} \Delta t & P_{ST,t} \le 0 \\ W_{ST,t} = W_{ST,t-1} - \dfrac{1}{\eta_{ST,out}} P_{ST,t} \Delta t & P_{ST,t} > 0 \end{cases} \tag{1}$$

From (1) one can state that energy losses are always present both in charging and discharging modes and this is taken into account introducing the efficiency of the storage unit. As one can notice, in principle, the efficiency in charging mode ($\eta_{ST,in}$) is different from the one in discharging mode ($\eta_{ST,out}$).

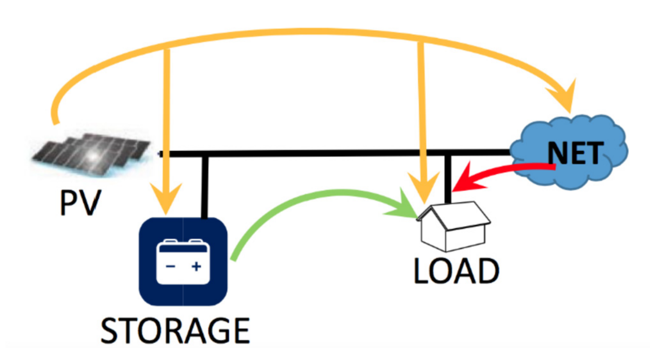


Fig. 1. Network and HRS scheme. The arrows show that, according to the HRS rules, the storage system can be charged only by the PV power plant and the electric load can be satisfied by the PV and/or the storage system and/or the external network.

A reasonable design criterion needs to include suitable constraints in order to avoid an unfeasible operational condition. The first one is the maximum and minimum SoC of the storage:

$$W_{ST,min} \le W_{ST,t} \le W_{ST,max} \tag{2}$$

A second constraint to be considered is the maximum power that the storage can deliver or absorb instantaneously, i.e.:

$$P_{ST,ch} \le P_{ST,t} \le P_{ST,dsch} \tag{3}$$

The network topology is neglected; as a consequence, one can suppose that both the load and the three energy sources (PV unit, storage device and external network) are connected to the same bus. Such hypothesis (i.e. the so-called single bus-bar (SBB) model [27]), limits the CPU effort and allows writing the power system balance as:

$$P_{NET,t} + P_{ST,t} + P_{PV,t} = P_{EL,t}. \tag{4}$$

From a mathematical point of view, (4) is an equation with two unknown quantities, since only the PV production and the load request are supposed to be known. This lead to the necessity of provide a suitable storage management in order to define the storage power production $P_{ST,t}$ to optimize the operation of the system. Since the aim of the article is that of providing an analytical and applicable procedure to size the PV and storage units, the storage management to be considered does not have to rely on complex EMS requiring the numerical solution of a constrained optimization problem. Instead, a simple solution to this problem is proposed, detailed in the next subsection.

2.3. Storage management strategy considered for the proposed integrated PV-Storage systems design

The problem posed at the end of the previous subsection requires a storage management strategy. In the present paper, a very simple but effective rule (the HRS) is used in order to avoid resorting to optimization problems; such rule exploits the fact that typically the cost of the purchased energy is much higher than the revenue of sold one. Consequently, economic optimization is achieved minimizing the power exchange with the external network. On this basis, one can apply the following rules:

1. if the renewable power generation is smaller than the electric demand ($P_{PV,t} < P_{EL,t}$), the system needs to discharge the battery until it is empty, (battery discharging mode);
2. if the renewable generation power is equal to demand ($P_{PV,t} = P_{EL,t}$), the storage should not provide or absorb any power;
3. if the renewable power generation is greater than the electric demand ($P_{PV,t} > P_{EL,t}$), the storage should charge until it is fully charged (battery charging mode);

For all three cases, one can calculate the storage energy set point $P_{ST,t}$ for the given time frame as:

$$P_{ST,t} = P_{EL,t} - P_{PV,t}. \tag{5}$$

In accordance with (2) and (3) the storage power production needs to respect two constraints: the first one is not exceeding the maximum power production in charging or discharging configuration. The second one is not choosing a power set point that would violate the maximum or minimum storage energy in the time interval $[t\Delta t, (t+1)\Delta t]$. This last constraint can be respected calculating the maximum or minimum power production imposing $W_{ST,t} = W_{ST,\min}$ or $W_{ST,t} = W_{ST,\max}$ in the first and second of (1) and solving it with respect to $P_{ST,t}$. In accordance to the sign of (5) one has that:

$$P_{ST,t} = \begin{cases} \min\left[P_{EL,t} - P_{PV,t}, P_{ST,disch}, (W_{ST,t-1} - W_{ST,\min})\eta_{ST,out} \big/ \Delta t\right] & (P_{EL,t} - P_{PV,t}) \geq 0 \\ \max\left[P_{EL,t} - P_{PV,t}, P_{ST,ch}, (W_{ST,t-1} - W_{ST,\max}) \big/ \left(\eta_{ST,in}\Delta t\right)\right] & (P_{EL,t} - P_{PV,t}) < 0 \end{cases} \quad (6)$$

Once that $P_{ST,t}$ is known, the actual energy content can be obtained using (1), while the power injected by the external network can be calculated inserting $P_{ST,t}$ into (4).

As a consequence, considering a set of N_t time intervals to define the time horizon (e.g. one day), one can set up the following procedure (also depicted as flowchart in Fig. 2):

1. Acquire the storage SoC $W_{ST,t}$, the PV production $P_{PV,t}$ and the electric load $P_{EL,t}$;
2. If $P_{PV,t} = P_{EL,t}$ then $P_{ST,t} = 0$ otherwise compute $P_{ST,t}$ according to (6);
3. Evaluate $P_{NET,t}$ using (4);
4. Evaluate $W_{ST,t+1}$ according to (1);
5. repeat the procedure until the end of the time horizon;

2.4. Design criterion for the integrated PV-Storage solution

Once the system structure and the management logic are defined, it is possible to proceed with the definition of a suitable criterion to size an integrated solution in self-consumption configuration. In this case, one has the necessity of defining a proper rating for both the PV unit and the storage device (i.e. $P_{PV,\max}$ and $W_{ST,\max}$). An optimal sizing of the integrated solution is, for sure, a trade-off between the system performance and the initial investment one has to undertake. In order to achieve it, one should consider:

1. an estimation of the load time profile during system expected lifetime (this would vary in accordance to the peculiarity of the load, residential, commercial, industrial, etc.);
2. an estimation of the expected PV power production time profile per kW of installed power (i.e. irradiance data of the site) for the system expected lifetime;
3. the storage management strategy.

The first two points can be addressed resorting to databases that provide a one-year day by day load power request [29] and solar irradiance time profile [30] (which can be converted into PV power production according to [31]). As far as the third issue is concerned, the HRS defined in the previous section is used as the base for the following dissertation. Based on the previously mentioned data, one has to define a cost function C that links the overall cost/gain provided by the integrated solution during its lifetime to the PV unit and storage device ratings ($P_{PV,max}$ and $W_{ST,max}$) and then evaluate the cost function for a predefined set of PV and storage sizes among the ones available on the market and choose the most advantageous combination (i.e. the one characterized by the minimum cost/maximum revenue).

The cost function has to encode the investment cost C_0 to purchase the storage and the PV systems and the variable cost/gain associated with the energy exchange with the external network during the system lifetime, $C_{NET,\ life}$.

The investment cost C_0 can be estimated according to the following considerations:

- the storage cost C_{ST} per installed kWh and the PV cost C_{PV} per installed peak kW are typically available data ([32] and [33]).
- the expected life of a PV system is about 20 years and the one of the storage device is about 10 years

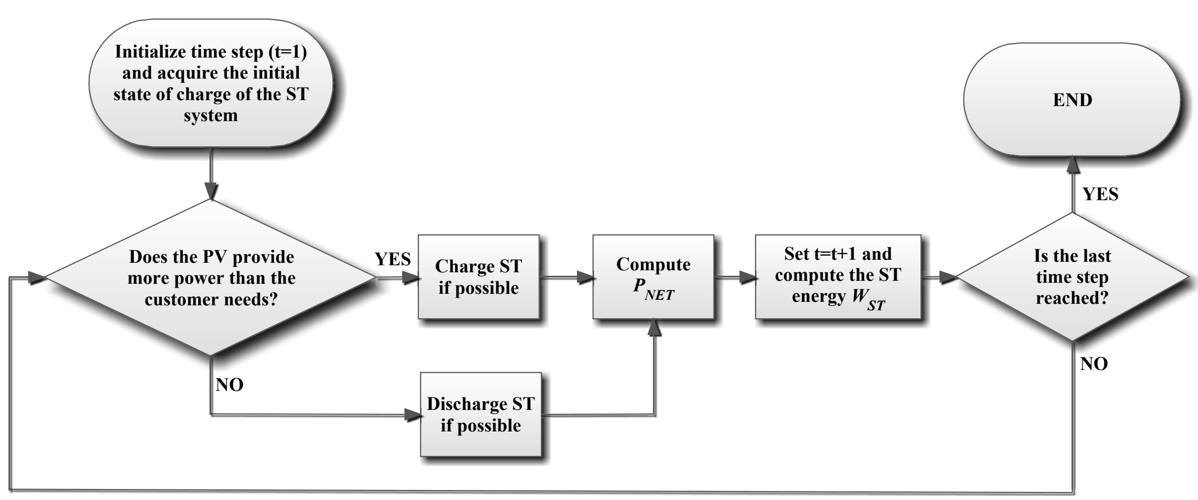


Fig. 2. Flow chart of the HRS.

On this basis one can write the investment cost C_0 as:

$$C_0(P_{PV,\max}, W_{ST,\max}) = C_{PV}P_{PV,\max} + C_{ST}W_{ST,\max} + \frac{C_{ST}W_{ST,\max}}{(1+WACC)^{9.5}} \tag{7}$$

which accounts also for the Weighted Average Cost of Capital ($WACC$) in the evaluation of the cost of the second storage device occurring after the first 10 years.

The cost/gain coming from the energy buying/selling from the Distribution System Operator (DSO) in the range $[t\Delta t, (t+1)\Delta t]$ can be evaluated according to the following relation:

$$C_{NET,t}(P_{PV,\max}, W_{ST,\max}) = \begin{cases} C_{sell,t}P_{NET,t}\Delta t & P_{NET,t} < 0 \\ C_{buy,t}P_{NET,t}\Delta t & P_{NET,t} \geq 0 \end{cases} \tag{8}$$

$C_{sell,t}$ and $C_{buy,t}$ being the energy selling and purchasing price respectively and $P_{NET,t}$ being calculated according to (4). The dependence of $C_{NET,t}$ on $P_{PV,max}$ and $W_{ST,max}$ is due to the fact that the power exchange with the external network depends on the power production of the PV unit that, at fixed irradiance, is directly proportional to the system rating.

So, choosing one year as the time horizon for the flowchart of Fig. 2, one has that the energy annual cost $C_{NET,Y,j}$ for the year j $(j = 1, \ldots, 20)$ is given by:

$$C_{NET,Y,j}(P_{PV,\max}, W_{ST,\max}) = \sum_{t=1+(j-1)N_t}^{jN_t} C_{NET,t}(P_{PV,\max}, W_{ST,\max}) \tag{9}$$

having chosen N_t such that $N_t\Delta t$ is one year. Finally, the total energy cost/gain is obtained summing $C_{NET,Y,j}$ over 20 years, again considering the Weighted Average Cost of Capital (WACC), as follows:

$$C_{NET,life}(P_{PV,\max}, W_{ST,\max}) = \sum_{j=1}^{20} \frac{C_{NET,Y,j}(P_{PV,\max}, W_{ST,\max})}{(1+WACC)^{j-0.5}} \tag{10}$$

Therefore, the function C is given by:

$$\begin{aligned} C(P_{PV,\max}, W_{ST,\max}) &= C_{NET,life}(P_{PV,\max}, W_{ST,\max}) \\ &\quad + C_0(P_{PV,\max}, W_{ST,\max}) = \\ &= \sum_{j=1}^{20} \frac{C_{NET,Y,j}(P_{PV,\max}, W_{ST,\max})}{(1+WACC)^{j-0.5}} \\ &\quad + \left[C_{PV}P_{PV,\max} + C_{ST}W_{ST,\max} + \frac{C_{ST}W_{ST,\max}}{(1+WACC)^{9.5}}\right]. \end{aligned} \tag{11}$$

In order to account for reasonable variation of the load and irradiance for the PV, that will not be the same for every year, the following corrective actions are implemented:

- a random error with upper limit equal to 10% should be added on the load request $P_{EL,t}$ of (4) to account for different load demands in different years;
- a suitable function [31] is introduced that models the fact that, starting from the same solar irradiance conditions, the PV power production $P_{PV,t}$ decreases year by year.

Finally, one can consider a set of N_{PV} PV plant ratings and a set of N_{ST} storage ratings selected among the sizes commercially available, evaluate (11) $N_{PV}N_{ST}$ times corresponding to any combination of $P_{PV,max,k}$ and $W_{ST,max,k}$ and find the minimum value assumed.

In the following $P^{opt}_{PV,\max}$ and $W^{opt}_{ST,\max}$ denotes the values of the chosen PV and storage sizes, and C^{opt} the corresponding value of (11).

3. Economic analysis

In order to demonstrate the economic sustainability of the proposed solution, an analysis based on the main investments evaluation criteria is necessary. For this reason, this section provides the theoretical evaluation of the most important parameters to be assessed in order to demonstrate the economic advantage of the proposed optimized solution with respect to a different one. The parameters taken into account are the Payback Period (*PBP*), the Net Present Value (*NPV*), the Discounted Profitability Ratio (*DPR*) and the Internal Rate of Return (*IRR*).

This analysis allows evaluating the proposed technological solution taking into account both the initial investment and future revenues/costs.

In principle, all the parameters are functions of the storage and PV plant rating, namely $P_{PV,\max}$ and $W_{ST,\max}$ and can be calculated for any choice of the component rating. However, in the following, the parameters are calculated corresponding to the optimal ratings defined in Section 2 ($P^{opt}_{PV,\max}$ and $W^{opt}_{ST,\max}$) to check whether this solution, evaluated minimizing the overall cost, is also efficient from the standpoint of all the other economic parameters.

3.1. Payback period

The *PBP* is the period of time required to recoup the funds expended in an investment. It can be evaluated using

$$PBP = \min\left\{n : \sum_{j=1}^{n} FCFO_j - FCFO_0 \geq 0\right\} \tag{12}$$

where $FCFO_j$ (with $j = 1, \ldots, 20$) is the cash flow from operations of year j corresponding to:

$$FCFO_j = \left[C_{NET,Y,j}(0,0) - C_{NET,Y,j}\left(P^{opt}_{PV,\max}, W^{opt}_{ST,\max}\right)\right] - \begin{cases} 0 & j \neq 10 \\ C_{ST}W^{opt}_{ST,\max} & j = 10 \end{cases} \tag{13}$$

while $FCFO_0$ is the cash flow from operations of year 0, representing the starting investment:

$$FCFO_0 = C_{PV}P^{opt}_{PV,\max} + C_{ST}W^{opt}_{ST,\max} \tag{14}$$

In the case study the solution of equation (12) can be obtained finding the first integer number n such that:

$$\sum_{j=1}^{n}\left[C_{NET,Y,j}(0,0) - C_{NET,Y,j}\left(P^{opt}_{PV,\max}, W^{opt}_{ST,\max}\right)\right] - \left[C_{PV}P_{PV,\max} + \begin{cases} C_{ST}W^{opt}_{ST,\max} & n < 10 \\ 2C_{ST}W^{opt}_{ST,\max} & n \geq 10 \end{cases}\right] \geq 0 \tag{15}$$

3.2. Net Present Value

The *NPV* measures the excess or shortfall of cash flows, in present value terms, above the cost of funds. It is defined as the algebraic sum of the $FCFO_j$ over all the years of the analysis horizon discounted at an interest rate ($WACC$). In the case study the NPV is:

$$NPV = \sum_{j=1}^{20} \frac{FCFO_j}{(1+WACC)^{j-0.5}} - FCFO_0$$
$$= \sum_{j=1}^{20} \frac{\left[-C_{NET,Y,j}\left(P^{opt}_{PV,\max}, W^{opt}_{ST,\max}\right) + C_{NET,Y,j}(0,0)\right]}{(1+WACC)^{j-0.5}} +$$
$$-\left[C_{PV}P^{opt}_{PV,\max} + C_{ST}W^{opt}_{ST,\max} + \frac{C_{ST}W^{opt}_{ST,\max}}{(1+WACC)^{9.5}}\right] \tag{16}$$

3.3. Discounted Profitability Ratio

The DPR provides the percentage return of the investment expenditure along the lifetime of the project. It is the ratio between the Net Present Value and the Initial Investment:

$$DPR(\%) = 100\frac{NPV}{FCFO_0} \tag{17}$$

3.4. Internal Rate of Return

The IRR is the rate of return that makes the NPV equal to zero. It can also be defined as the discount rate at which the present value of all future cash flow is equal to the initial investment or in other words the rate at which an investment breaks even.

IRR calculations are commonly used to evaluate the desirability of investments or projects. The higher a project's IRR, the more desirable the project. It can be calculated, starting from (16), as:

$$\sum_{j=1}^{20} \frac{FCFO_j}{(1+IRR)^{j-0.5}} - FCFO_0 = 0 \tag{18}$$

As a general comment, none of these parameters can be sufficient for an exhaustive evaluation of the considered solution if taken one by one. Only an integrated analysis can give a complete and appropriate investment evaluation.

4. Numerical results

This section is dedicated to the numerical validation of the proposed method for the sizing of the integrated PV-Storage solution. It is divided in two steps: the first one provides a validation of the proposed HRS in comparison with a conventional EMS that relies on an optimization algorithm (in order to validate if the hypothesis of the proposed procedure are well posed); the second is the actual verification of the proposed optimal sizing strategy starting from the knowledge of the solar irradiance and the electric load power profiles. The storage numerical data used for the proposed validation are reported in Table 1.

Table 1
Storage numerical data.

Mean performance coefficients $\eta_{ST,in}$	0.9
Mean performance coefficients $\eta_{ST,out}$	0.9
Maximum charging power $P_{ST,ch}$	$-(0.3\ \mathrm{h}^{-1})W_{ST,\max}$
Maximum discharging power $P_{ST,dsch}$	$(0.3\ \mathrm{h}^{-1})W_{ST,\max}$
Depth of Discharge (DoD)	0%

Moreover, the cost of the storage system is quantified at 350 \$/kWh [32] and the cost of the PV power plant at 3000 \$/kW$_p$ for residential users and 1800 \$/kW$_p$ for commercial ones [33].

Finally, the energy selling price C_{sell} is equal to 0.05 \$/kWh ([33] and [34]) and the purchase one C_{buy} is equal to 0.17 \$/kWh [35], while the possible ratings of the PV and storage are chosen from a discrete set with steps of 0.25 kW and 1 kWh, respectively. For the economic analysis, $WACC$ is equal to 0.035.

4.1. Validation of the HRS

The aim of the present subsection is to validate the presented storage system charging/discharging strategy. To do this, a commercial user (restaurant) located in Connecticut has been chosen and the corresponding load and solar irradiance time profiles have been picked up from Refs. [29] and [30], respectively. Such user is supposed to be characterized by 200 kWh of storage system and 150 kW of PV unit; then, the HRS has been applied for one year and its results have been compared with the ones provided by the EMS proposed in Refs. [27] and [28] which relies on an optimization algorithm. Fig. 3 depicts the load request and the PV units production time profiles for one year, putting in the horizontal axis the days of the year and in the vertical axis the hours of the day.

Fig. 4 plots the storage power production/absorption and the power exchange with the external network as dictated by the HRS and with the same format as Fig. 3, while the EMS solution is represented in Fig. 5. As one can see, there are very small differences between the two approaches. A more quantitative comparison is then performed in Fig. 6, where the monthly cost of the energy exchanged with the public grid is reported as the result of the HRS (red) and the EMS (yellow). Examining Fig. 6, it is apparent

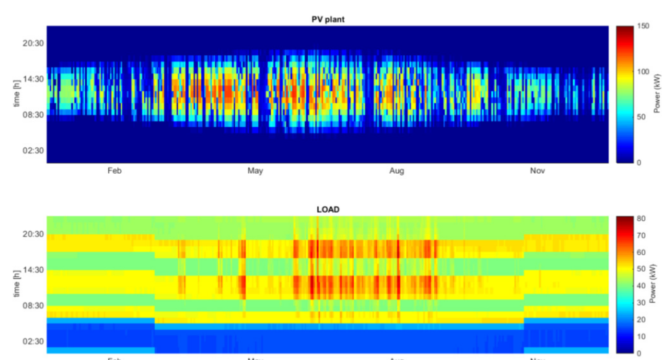


Fig. 3. Load request and PV unit production time profiles.

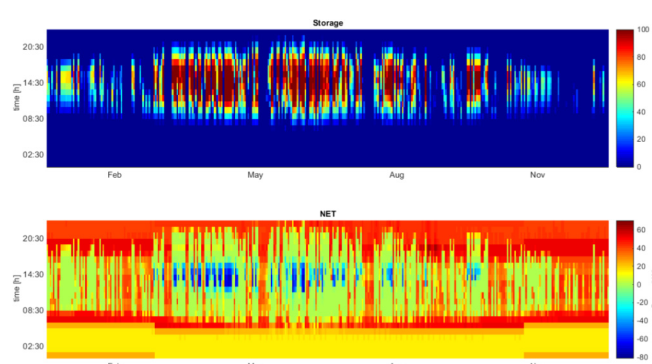


Fig. 4. HRS power profile for the Storage System and the External Network for the Commercial Restaurant test case during the year.

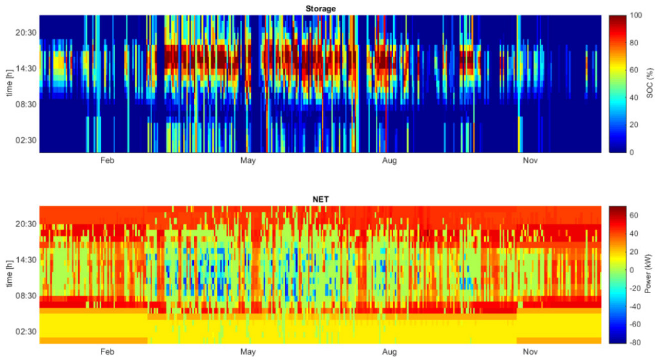


Fig. 5. EMS power profile for the Storage System and the External Network for the Commercial Restaurant test case during the year.

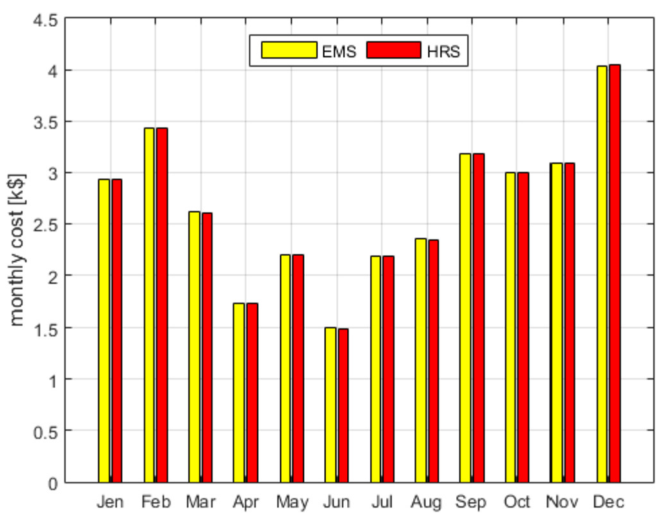


Fig. 6. Monthly energy cost as dictated by the HRS (red) and the EMS (yellow).

that there is an excellent agreement between the two approaches, which shows the good performances of the proposed battery management strategy with the advantage that it is very simple to be implemented in the evaluation of the total investment cost as detailed in Section 2 without the usage of an optimization solver which is seldom free, especially for industrial applications.

4.2. Validation of the proposed rating strategy for the PV storage integrated solution

The second part of this section is relevant to the application of the proposed procedure to find out a proper rating for the PV and the storage systems in different application cases. In order to cover a wide possibility of application cases, residential and commercial users are analysed. Residential test cases are dived in three scenarios, namely low, medium and high load, while commercial ones are differentiated considering a large hotel, a restaurant and a supermarket.

The results of the proposed sizing strategy for the three residential cases are reported in Table 2 including the optimized cost coming from (11) and all the economic indexes cited in Section 3.

As one can see, the PV rating increases along with the load request; this sounds reasonable because, as the selling energy price is much smaller than the buying one, the best situation is the one in which the load is completely fed by the PV unit. In this case, the relevant cost of the storage unit makes it inconvenient to be installed.

Table 2
Overall cost, chosen size and the corresponding economic parameters for the Residential loads.

	Residential low	Residential base	Residential high
$P^{opt}_{PV,\max}$ [kW]	1.00	2.00	3.00
$W^{opt}_{ST,\max}$ [kWh]	0.00	0.00	0.00
C^{opt} [k$]	21.77	46.99	75.63
PBP [year]	13.00	13.00	13.00
NPV [k$]	0.29	0.63	0.91
DPR [%]	11.77	12.67	12.19
IRR [%]	4.84	4.94	4.89

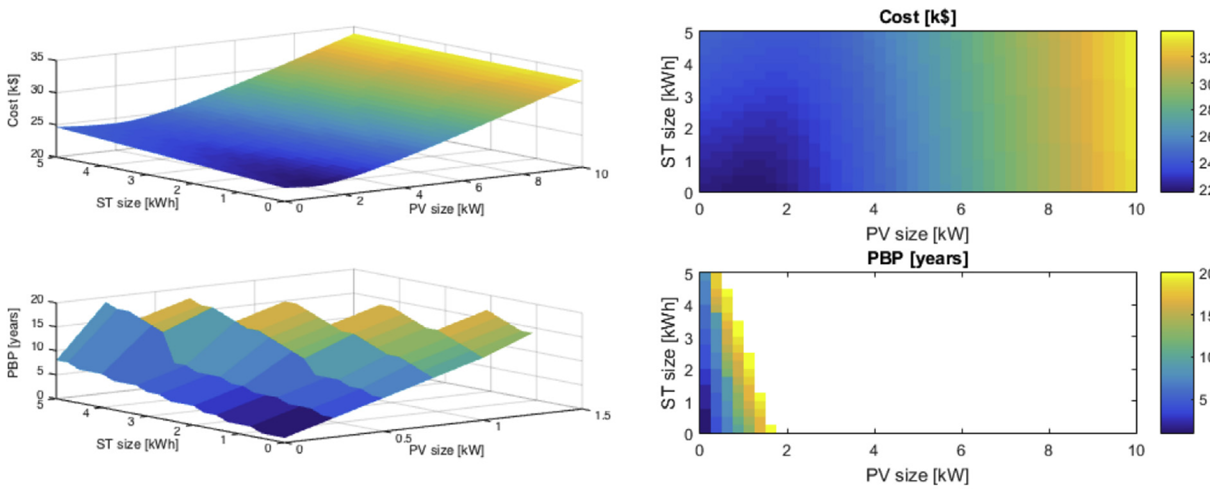
As far as the economic parameters are concerned, the *PBP* is the same for all the three cases, so there is no difference in the capability to cover the initial investment. The same is for the *IRR*, which is always around 5%, guaranteeing the same (low) safety level against *WACC* fluctuations for all the configurations. The *NPV* value, instead, significantly increases with the load level, for the same reason the PV rating does. The *DPR* value varies in a less sensitive way due to the proportional increase of the *NPV* and of the initial investment. By comparing the three cases, the most profitable one appears to be the one with higher load even if the *IRR* low level makes all the cases quite risky.

The sensitivity of the total cost (11) and the *PBP* as functions of the PV and ST sizes is represented in Figs. 7, 8 and 9 for the three residential test cases. It can be observed that, for the chosen range of storage and PV sizes, the total cost varies more in the case of low load (its maximum is more than 50% its minimum) rather than for the high load case (its maximum is less than 7% higher than its minimum). Moreover, the PBP is more sensitive to the PV size than to the storage one.

The results of the commercial test cases are reported in Table 3 using the same structure as Table 2. As for the residential test cases, the same considerations can be drawn for the rating of the PV system (increasing with the load) and for the absence of storage power. This once again implies that the relevant cost of the storage does not satisfy an optimized economic criterion. As far as the economic indexes are concerned, it is shown that the *PBP*, *IRR* and *DPR* are again equal in all the three cases. The *NPV* value, instead, is significantly higher in the Large Hotel test case. By comparing the three cases, it is clear that the most profitable one appears to be this latter. If this investment solution is compared to the residential best case (residential with high load) it is possible to affirm that the commercial investment is more profitable in terms of generated cash flow both in absolute and percentage terms on the investment values and it is also safer against possible *WACC* fluctuations because the *IRR* value is higher (more than 3% with respect to the previous situation).

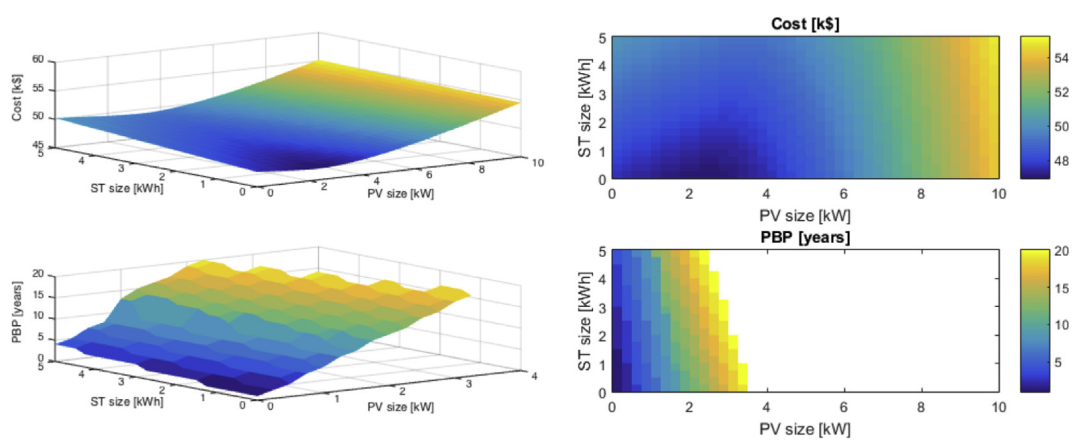
The sensitivity of the total cost (11) and the *PBP* as functions of the PV and storage sizes is represented in Figs. 10, 11 and 12 for the three commercial test cases. It can be observed once again that, in the chosen range of storage and PV sizes, the total cost has a higher variation for smaller loads (the Restaurant) as its maximum is more than 200% its minimum if compared to the highest one (large Hotel test case), where its maximum is less than 9% its minimum. Also for this set of test cases, the PBP is more sensitive to the PV size than to the storage one.

In the light of the results obtained for the six considered test cases, pointing out the negative impact of the introduction of the storage system in combination with the PV one, the proposed algorithm has been used to extrapolate how much the storage



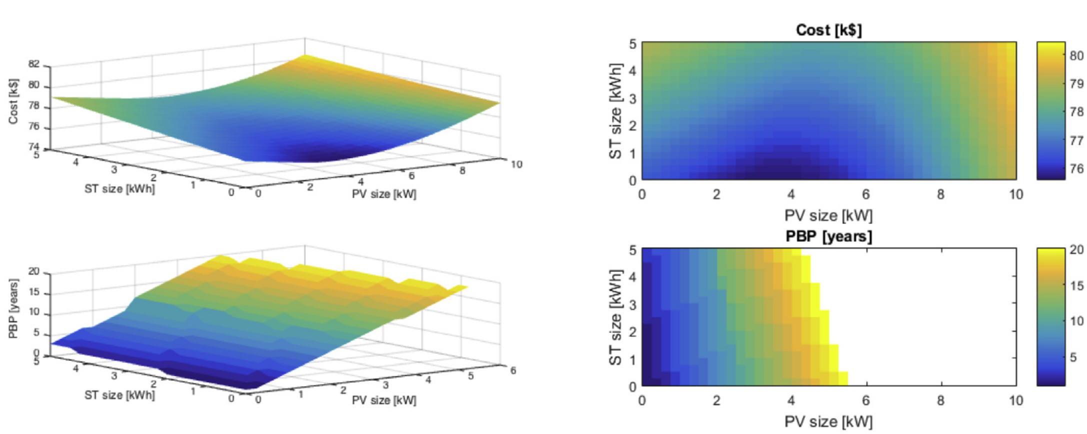
(a) (b)

Fig. 7. Cost and PBP values as functions of PV and storage rating in the case of low residential load (a) 3D plot, (b) colour level 2D plot. The white region in the PBP (b) graph means that the index is greater than 20 years.



(a) (b)

Fig. 8. Cost and PBP values as functions of PV and storage sizes in the case of base residential load (a) 3D plot, (b) colour level 2D plot. The white region in the PBP graph (b) means that the index is greater than 20 years.



(a) (b)

Fig. 9. Cost and PBP values as functions of PV and storage sizes in the case of high residential load (a) 3D plot, (b) colour level 2D plot. The white region in the PBP graph (b) means that the index is greater than 20 years.

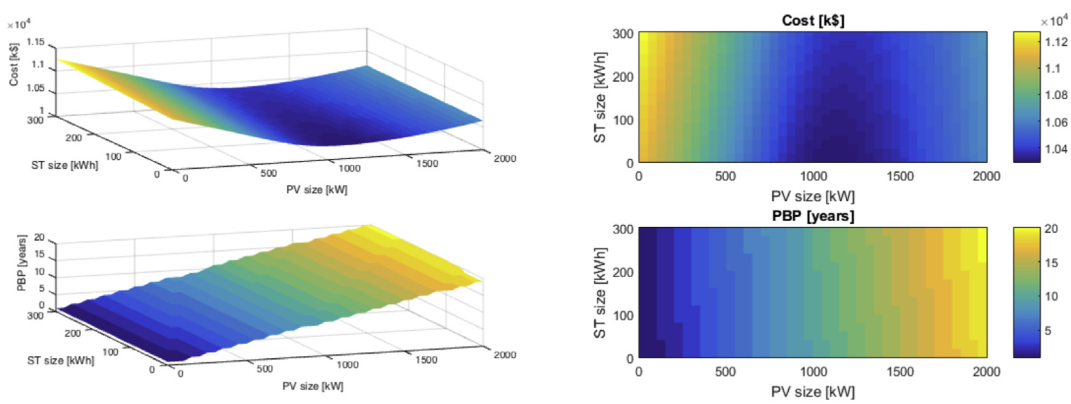
Table 3
Results of the proposed sizing strategy for the commercial test cases.

	Large hotel	Restaurant	Supermarket
$P^{opt}_{PV,\max}$ [kW]	1100.00	100.00	760.00
$W^{opt}_{ST,\max}$ [kWh]	0.00	0.00	0.00
C^{opt} [k$]	10294.22	789.88	5123.06
PBP [year]	11.00	11.00	11.00
NPV [k$]	819.49	73.71	580.22
DPR [%]	39.63	39.21	40.61
IRR [%]	7.79	7.75	7.89

unitary purchasing price should be reduced in order to make a positive impact on the integrated system economic asset. In order to do so, the same test cases previously considered are processed for increasing values of the storage purchase price starting from a lower limit value of 50 $/kWh. The resulting storage ratings are reported in Table 4. An analysis of the results achieved suggests that for very large loads (the large Hotel and restaurant test cases) a threshold value of 150 $/kWh can be identified, which is encouraging as the storage price is forecast to reach this value in the near future [32]. For residential application, on the other hands, the proposed analysis indicates that the installation of the integrated PV-Storage solution is far from being competitive.

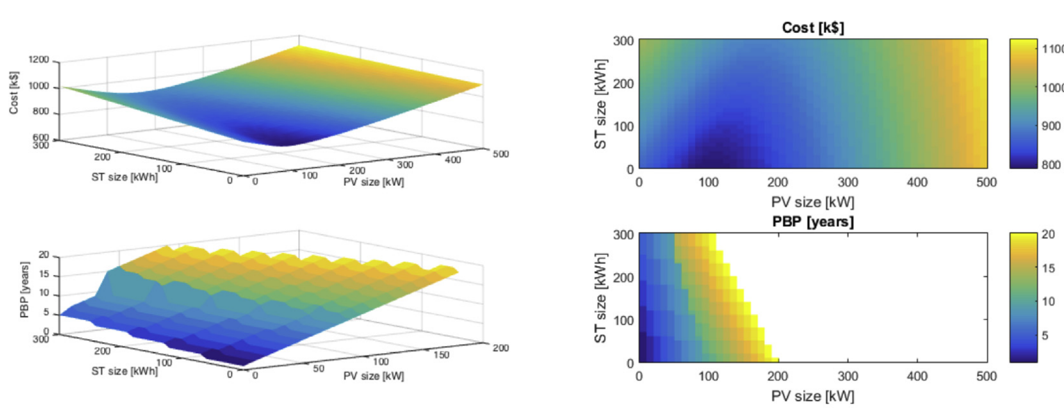
5. Conclusions

The present article proposed an analytical approach to design an integrated PV-Storage solution for self-consumption purposes optimizing the overall system cost. In order to define this strategy, first a simple but effective logic to manage the charging and discharging phases of the storage devices has been defined, avoiding the usage CPU demanding optimization problems. Then, the paper has detailed an analytical procedure to evaluate the overall cost of the integrated solution accounting for the initial investment, operational costs and revenue and the different lifetime of the storage and PV systems. Subsequently, the use of some classical economic indexes has been proposed in order to compare different solutions from an economic point of view. Numerical simulations have been performed in various test cases first with the aim of validating the reliability of the implemented simplified management logic for the storage system, highlighting an excellent agreement with a classical EMS and reduced computational costs. This second part of the validation has been performed in six different test cases starting from reasonable (commercial and residential) loads and solar irradiance profiles. Numerical results pointed out that the current price of storage units is still too high to justify its integration with PV system from a purely economic point of view. Moreover, the complete economic



(*a*) (*b*)

Fig. 10. Cost and PBP values as functions of PV and storage sizes in the case of large hotel (a) 3D plot, (b) colour level 2D plot.



(*a*) (*b*)

Fig. 11. Cost and PBP values as functions of PV and storage sizes in the case of restaurant (a) 3D plot, (b) colour level 2D plot. The white region in the PBP graph (b) means that it is greater than 20 years.

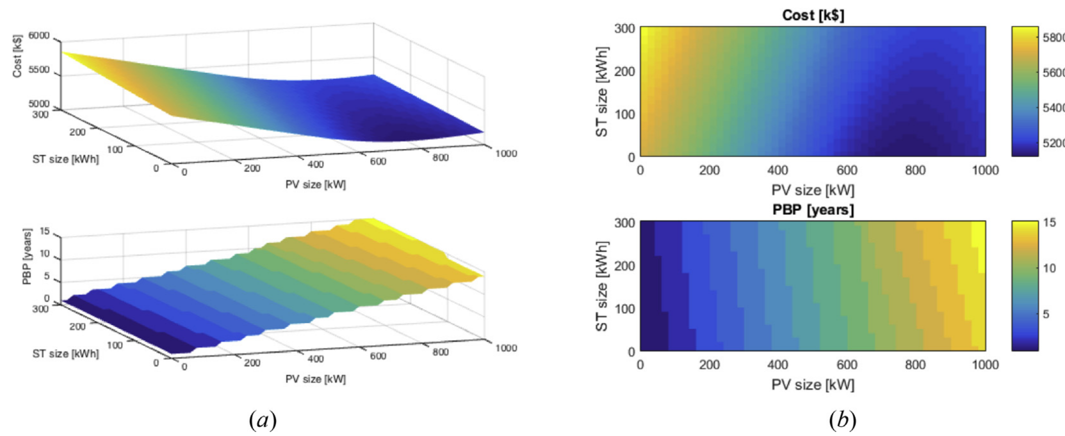


Fig. 12. Cost and PBP values as functions of PV and storage sizes in the case of supermarket (a) 3D plot, (b) colour level 2D plot. The white region in the PBP graph (b) means that it is greater than 20 years.

Table 4
Storage system rating as a function of the cost per kWh.

	Residential low	Residential base	Residential high	Large hotel	Restaurant	Supermarket
50 $/kWh	0 kWh	0 kWh	1 kWh	2100 kWh	410 kWh	2100 kWh
100 $/kWh	0 kWh	0 kWh	0 kWh	1070 kWh	200 kWh	400 kWh
150 $/kWh	0 kWh	0 kWh	0 kWh	100 kWh	100 kWh	0 kWh
200 $/kWh	0 kWh	0 kWh	0 kWh	0 kWh	0 kWh	0 kWh
250 $/kWh	0 kWh	0 kWh	0 kWh	0 kWh	0 kWh	0 kWh
300 $/kWh	0 kWh	0 kWh	0 kWh	0 kWh	0 kWh	0 kWh

analysis has shown that the commercial investment solution is the most economically advantageous, because it ensures a good protection level against possible *WACC* fluctuations and a high profitability level, both in terms of generated cash flow and ratio between new cash flow and initial investment.

Finally, a backward application of the proposed approach has been used to determine how much the storage purchasing price will have to reduce in order to make the integrated PV solution economically sustainable, highlighting the possibility of reaching the necessary storage price in a reasonable short time.

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